

EPIC

Epic Card

Title

Build and Automate CI Workflow for Healthcare Appointment Management Platform

Description

Develop healthcare appointment APIs, automate testing, manage repository, and implement CI pipeline using GitHub Actions.

FEATURES

Create these cards inside the **Features** list.

Feature 1 — Appointment API Development

Description

Develop APIs for healthcare appointment management services.

Feature 2 — Doctor Availability Tracking

Description

Develop functionality to verify doctor availability.

Feature 3 — Patient Appointment Validation

Description

Validate and display patient appointment confirmation status.

Feature 4 — Repository Management

Description

Maintain application source code using GitHub version control.

Feature 5 — Automated Testing

Description

Create automated API testing using pytest.

Feature 6 — CI Pipeline Setup

Description

Configure GitHub Actions CI workflow automation.

Feature 7 — Build Verification

Description

Validate pipeline execution and successful build completion.

USER STORIES

Create these cards inside the **User Stories** list.

User Story 1 — Appointment Status

As a patient, I want to check appointment status so that I know whether my appointment is confirmed.

User Story 2 — Doctor Availability

As a healthcare admin, I want to verify doctor availability so that appointments can be scheduled properly.

User Story 3 — Repository Management

As a DevOps engineer, I want code stored in GitHub so that version control is maintained.

User Story 4 — Automated Testing

As a tester, I want automated test execution so that application quality is validated.

User Story 5 — CI Pipeline Automation

As a DevOps engineer, I want GitHub Actions pipeline automation so that builds run automatically after code push.

User Story 6 — Dependency Installation

As a developer, I want dependency installation automated so that builds are consistent.

User Story 7 — Pipeline Verification

As a project team member, I want pipeline verification so that deployment readiness is ensured.

TASKS

Create these cards inside the **Tasks** list.

Task 1 — Create Flask Application

Checklist

- Create app.py
 - Import Flask
 - Create Flask application object
 - Create home API
 - Create health API
 - Create appointment-status API
 - Create doctor-availability API
 - Run Flask server
-

Task 2 — Create Requirements File

Checklist

- Create requirements.txt
 - Add flask dependency
 - Add pytest dependency
-

Task 3 — Create Unit Tests

Checklist

- Create test_app.py
 - Test home API
 - Test health API
 - Test appointment-status API
 - Test doctor-availability API
-

Task 4 — Configure Git Repository

Checklist

- Initialize git repository
 - Add project files
 - Commit code
 - Create GitHub repository
 - Connect remote repository
 - Push code to GitHub
-

Task 5 — Create GitHub Actions Workflow

Checklist

- Create .github folder
 - Create workflows folder
 - Create python-app.yml
 - Configure workflow trigger
 - Configure Python setup
 - Install dependencies
 - Run pytest tests
-

Task 6 — Validate CI Pipeline

Checklist

- Push latest code
- Open GitHub Actions tab
- Verify workflow execution
- Verify all tests passed
- Verify successful build